

Translated

From: e-maxx.ru

Newton's method for finding roots

This is an iterative method invented by Isaac Newton around 1664. However, this method is also sometimes called the Raphson method, since Raphson invented the same algorithm a few years after Newton, but his article was published much earlier.

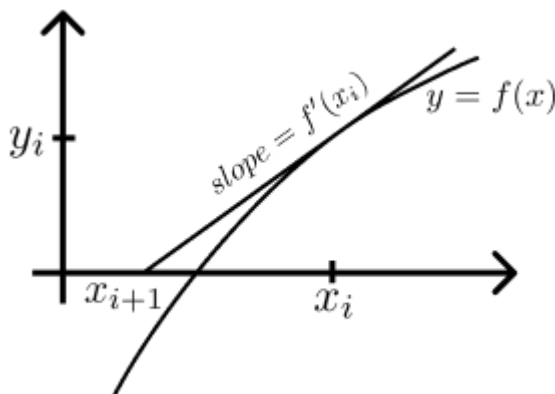
The task is as follows. Given the following equation:

$$f(x) = 0$$

We want to solve the equation. More precisely, we want to find one of its roots (it is assumed that the root exists). It is assumed that $f(x)$ is continuous and differentiable on an interval $[a, b]$.

Algorithm

The input parameters of the algorithm consist of not only the function $f(x)$ but also the initial approximation - some x_0 , with which the algorithm starts.



Suppose we have already calculated x_i , calculate x_{i+1} as follows. Draw the tangent to the graph of the function $f(x)$ at the point $x = x_i$, and find the point of intersection of this tangent with the x -axis. x_{i+1} is set equal to the x -coordinate of the point found, and we repeat the whole process from the beginning.

It is not difficult to obtain the following formula,